Performance of the Veterans Affairs Low Vision Visual Functioning Questionnaire

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Abstract: The purpose of this study was to develop an instrument that would be sensitive to the visual difficulties of persons with low vision and that could be used to evaluate outcomes and plan vision rehabilitation services. Toward that end, the Veterans Affairs Low Vision Visual Functioning Questionnaire was administered to 117 participants in rehabilitation programs in Veterans Affairs medical centers and in the private sector. The results demonstrate that the questionnaire is an effective instrument for measuring vision difficulties in persons who receive low-vision rehabilitation.

Approximately 240,000 new cases of visual impairments (that is, low vision and blindness) occur each year in the United States, and this number is expected to double over the next 25 years, according to estimates cited in an epidemiological review by Massof (2002). Given that most of the diseases and conditions that result in visual impairments are more prevalent in older people, the aging of baby boomers will strongly contribute to this increased number of cases. The rehabilitation community has been faced with the challenge of developing better methods and models for extending services to provide for the impending epidemic of vision loss (Massof, 1998). As Congress debates an increase in Medicare coverage for rehabilitation services (Capuano, 2000), rehabilitation scientists are faced with the task of evaluating methods and determining their effectiveness for improving the quality of life of persons with low vision (Stelmack, 2001).

A number of questionnaires have been developed to assess the quality of life of or the ability to perform daily tasks by individuals who have vision problems (for a review, see Massof & Rubin, 2001). However, the main emphasis of the design of the items and their scales has been on evaluating visual function in clinical tri-

An earlier version of this article was presented, in part, at the annual meeting of the Association for Research in Vision and Ophthalmology, Fort Lauderdale, FL, May 2001. The research on which this article was based was supported by a research grant from the U.S. Department of Veterans Affairs, Veterans Health Administration, and Rehabilitation Research and Development Service, Washington, DC; National Institutes of Health Core Grant EY01792, Bethesda, MD; and an unrestricted grant from Research to Prevent Blindness, New York, NY.

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als for specific treatable eye conditions, such as cataract (Mangione et al., 1992). Most measures have been validated against traditional measures of visual function, including visual acuity and visual fields, using correlation analyses that have yielded acceptable but not high correlations (Haymes, Johnston, & Heyes, 2000). The correlations are not expected to be high. As has been pointed out (Massof & Rubin, 2001), redundancy should be avoided, and the participants' self-report measures should provide wider profiles of functional performance than can be gained from laboratory tests of letter recognition and spot-luminance detection.

The aim of the research reported here was to use this effective means of obtaining more information from people with visual impairments and apply it to persons with low vision who are undergoing vision rehabilitation. The intent was to build a scale of items that are directly addressed by, and affected positively by, rehabilitation strategies. The methods used in this study were different from those used in previous work with low-vision participants in that we maintained a strong clinical orientation to vision rehabilitation. We were interested in developing a scale with clearly written questions about the performance of daily tasks that participants could easily understand. Another goal was to incorporate subcomponents of the items related to (1) whether any difficulty was the result of visual problems, (2) whether the participant was interested in receiving training for that activity, and (3) how the participant performed the activity. Even if an activity or task is difficult to perform, if the participant is not interested in training in it, then the activity or task becomes much less important for developing the participant's rehabilitation curriculum.

In addition, we used the Rasch rating-scale model (Wright & Masters, 1982) to construct a yardstick from the ordinal data provided by the questionnaire. The Rasch technique is optimal for developing an understandable ranking of tasks (for example, from the least to the most difficult). It has been used successfully in the development of interval scales, such as difficulty in assessing the activity levels of people who are blind (Becker, Lambert, Schulz, Wright, & Burnet, 1985; Schulz, Lambert, & Becker, 1985), the importance of accomplishing daily goals by persons with low vision (Massof, 1998), and the self-perceived ability for independent mobility in those with vision loss that is due to the inherited retinal degenerative disease retinitis pigmentosa (RP) (Turano, Geruschat, Stahl, & Massof, 1999). The ranking represents a useful yardstick to gauge a participant's functional growth after a rehabilitation program.

A pilot study (Stelmack, Szlyk, Joslin, Swetland, & Myers, 2000), in which the authors applied the National Eye Institute Visual Functioning Questionnaire-25 (NEI VFQ-25, 1996) in a Veterans Affairs (VA) setting, demonstrated some limitations in its use with persons who are undergoing vision rehabilitation. In this earlier work, Stelmack et al. found that 77 legally blind veterans from the Edward Hines VA Blind Rehabilitation Center, an inpatient program, reported statistically significant improvement on 30 of 39 items. The 9 items on which there was no significant improvement dealt with overall health, ocular pain, driving, socializing, and grooming. Stelmack et al. concluded that because the inpatient program does

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not focus on the rehabilitation of overall health or ocular pain, a change on these indexes would not be expected. In addition, the severity of the participants' vision losses precluded any training in driving. Another previous study in the senior author's laboratory showed no relationship between vision and either socializing or grooming (Szlyk et al., 2001).

Furthermore, Stelmack et al. (2000) found that almost one-fourth of the questions on the NEI VFQ-25 were not relevant to the study group. The also found that the older participants were confused about the directions and the wording of some of the items and whether the questions should be answered according to their corrected (eyeglass or contact lens) visual acuity or their visual acuity with low-vision devices. The goal became to develop a visual functioning questionnaire that would be appropriate for the aging participant of a low-vision rehabilitation program that could be used to design the participant's program and could provide a standard to measure progress.

This article reports the results from the baseline testing of participants who were enrolled in vision rehabilitation programs within the VA or in private-sector clinics. It includes the psychometric properties associated with each scale.

Methods

Participants

The Veterans Affairs Low Vision Visual Functioning Questionnaire was administered to 117 participants (97 men and 20 women), who ranged in age from 29 to 93 (mean age = 72 years). The participants were recruited from four rehabilitation programs: (1) the Visual Impairment Center to Optimize Remaining Sight (VICTORS) Program at the Chicago VA Health Care System, West Side Division, a short-term inpatient regional program that serves veterans who have less-severe vision loss than legal blindness (that is, a visual acuity of 20/200 or worse and/or a visual field diameter of 20 degrees or less) (20 participants), with visual acuity typically ranging from 20/40 to 20/200; (2) the Hines Blind Rehabilitation Center, described earlier (49 participants), with participants whose visual acuities are typically 20/200 and worse; (3) the Edwin and Lois Deicke Eye Center, a nonprofit agency providing low-vision services in the private sector (32 participants), serving participants with a spectrum of visual acuities from 20/40 and worse; and (4) the Massachusetts Eye and Ear Infirmary, a hospital-based rehabilitation program in Boston (16 participants), serving participants with a range of visual acuities that is similar to the Deicke Eye Center program. All the participants gave their informed consent to participate. The protocol was approved by the VA and University of Illinois at Chicago Institutional Review Boards.

Procedures

The questionnaire consisted of 48 items that evaluated functional performance in areas encompassing navigation, resolution, peripheral detection, finding, and general function (see Box 1). The questions were derived from items that had been successfully used in previous questionnaires and assessment batteries developed by the authors (Ross, Stelmack, Stelmack, & Fraim, 1991; Szlyk et al., 1997; Szlyk, Fishman, Grover, Revelins, & Derlacki, 1998; Szlyk et al., 2001), as well as

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Items on the Veterans Affairs Low Vision Visual Functioning Questionnaire

Is it difficult to . . .

1. Get dressed?
2. Keep clean?
3. Identify medicine?
4. Tell time?
5. Identify money?
6. Match clothes?
7. Groom yourself?
8. Identify food on a plate?
9. Eat and drink neatly?
10. Fix a snack?
11. Prepare meals?
12. Use appliance dials?
13. Clean the house?
14. Handle finances?
15. Make out a check?
16. Take a message?
17. Find something on a crowded shelf?
18. Find public restrooms?
19. Get around indoors in places you know?
20. Get around outdoors in places you know?
21. Get around in unfamiliar places?
22. Go out at night?
23. Go down steps in dim light?
24. Adjust to bright light?
25. Cross streets at traffic lights?
26. Use public transportation?
27. Get around in a crowd?
28. Avoid bumping into things?
29. Recognize persons up close?
30. Recognize persons from across the room?
31. Read street signs and store names?
32. Read headlines?
33. Read menus?
34. Read newspaper or magazine articles?
35. Read mail?
36. Read small print on package labels?
37. Read print on the television?
38. Keep your place while reading?
39. Watch television?
40. Play table and card games?
41. See photos?
42. Work on your favorite hobby?
43. Go to the movies?
44. Go to spectator events?
45. Play sports?
46. Do yard work?
47. Sign your name?
48. Read signs?

Box 1.

from interviews with rehabilitation specialists about the tasks that are taught in programs and with study participants regarding activities that they find most relevant to their lives.

There were four (*a–d*) subcomponents to each question. The first subcomponent (*a*) asked the participant to rate his or her difficulty with a particular task. The rating scale consisted of six response categories: (1) "not difficult," (2) "slightly difficult," (3) "moderately difficult," (4) "extremely difficult," (5) "impossible," and (6) "do not do it for nonvisual reasons." If the participant responded "1," the interviewer would skip to the *d* subcomponent of the question. If the participant responded "6," the interviewer would skip to the next question. The second subcomponent (*b*) asked the participant if any difficulty expressed in subcomponent *a* was because of her or his vision; the participant responded yes or no. If the participant responded no, the interviewer would skip to the next question.

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The third subcomponent (*c*) asked the participant if he or she wanted training in the activity; the participant responded yes or no. The fourth subcomponent (*d*) asked the participant how he or she usually performed the task. There were five response categories for subcomponent *d*: (1) "with my own eyes," (2) "with vision devices/techniques" (a magnifier), (3) "with my other senses or nonvision devices" (for example, a cane), (4) "someone helps me," and (5) "not applicable."

The questionnaire was administered to participants by telephone before their admission to the rehabilitation programs. The interviewer received consent from the participants by telephone, and the participants signed the consent form when they were admitted to the rehabilitation programs. As was stated earlier, data were analyzed with a Rasch rating-scale model (Wright & Masters, 1982), using WINSTEPS Version 3.16 Analysis Software. Correlational analyses were then used to relate the transformed Rasch measures.

Results

The scales

Figure 1 shows the difficulty scale in performing a task (questionnaire item) for subcomponent *a*, generated from the Rasch analysis, which gives the distribution of responses for items that were rated not difficult to impossible. The questionnaire items that even the most-able participants rated difficult (with a mean score of approximately 4, or extremely difficult) were reading small print on package labels, playing sports, reading newspapers or magazine articles, reading television print, reading menus, working on a favorite hobby, reading street signs and names of stores, recognizing persons from across the room, and reading mail. The items that even the least-able participants found not difficult (with a mean score of approximately 1) were keeping clean, getting around indoors in places you know, getting dressed, fixing a snack, eating and drinking neatly, getting around outdoors in places you know, grooming oneself, and telling time.

In Figure 1, the 48 items on the *y*-axis range from items that the 117 participants found the least difficult to do (top) to those they found the most difficult to do (bottom). The response distribution, ranging from scores for the most-able to the least-able participants, is shown on the *x*-axis, with the mean scores for each item indicated by the vertical line (M). As the cascade of numbers descends to the left, it indicates items that even the most-able participants found difficult. The Rasch analysis shows the full range of scores, but only bold-face numbers are meaningful. A response of 1 indicates not difficult, 2 indicates slightly difficult, 3 indicates moderately difficult, 4 indicates extremely difficult, and 5 indicates impossible. Colons indicate half-score points. Horizontal lines designate groupings of items that share the same response categories for the mean response.

The item ordering for the difficulty question (subcomponent *a*) was highly correlated with the item ordering for whether difficulty was related to vision (subcomponent *b*; $R = .87$, $p \leq .01$). As Figure 2 indicates, the participants reported most often that their difficulty reading small print on package labels, reading street signs and store names, reading newspapers or magazine articles, and reading menus was because of their vision loss. They reported least often that

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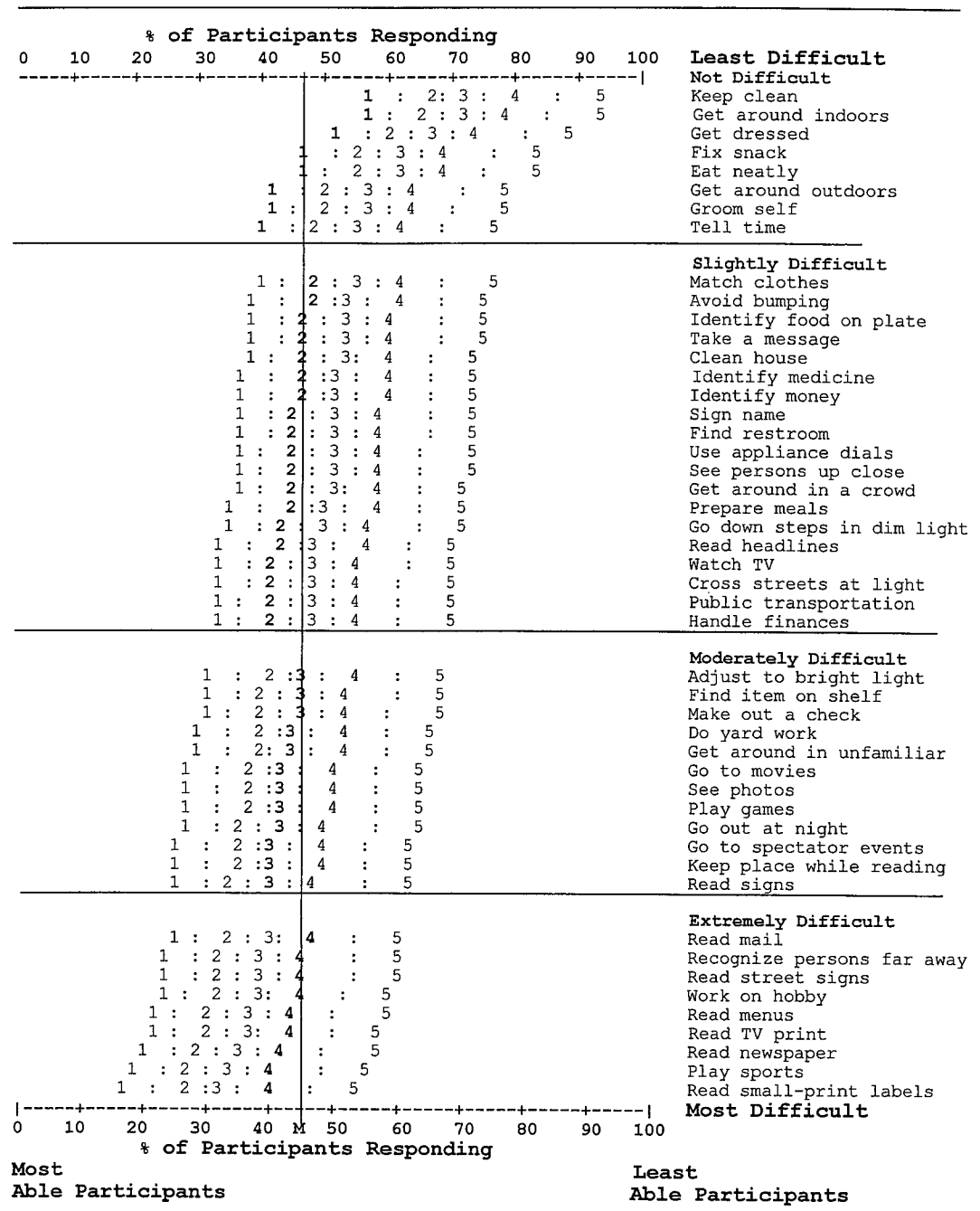


Figure 1. Ranking of responses to questionnaire subcomponent *a*: "Is it difficult to . . . ?"

any difficulty with such activities as keeping clean, getting around indoors in places you know, and getting dressed was because of their vision loss.

Note that the 48 items on the *y*-axis of Figure 2 range from items that the 117 participants least often found difficult because of their vision (top) to those they most often found difficult (bottom). A no, "Not because of my vision," response was coded 2. The most probable modal response is indicated by the *x*-axis. As the cascade of numbers descends to the left, it indicates items that even the most-able participants found difficult because of their vision. The vertical line (M) indicates the mean.

Also, the item ordering for the difficulty question (subcomponent *a*) was highly correlated with the item ordering for subcomponent *c* on whether the participants wanted training ($R = .85$, $p \leq .01$). Figure 3 shows the item orderings for subcomponent *c*. The 48 items on the *y*-axis range from those for which the participants had the least interest in receiving training (top) to those for which the participants had the greatest interest in receiving training (bottom). A no, "I do not want training," response was coded 2. The most probable modal response is indicated by the *x*-axis. As the cascade of numbers descends to the left, it indicates items for which even the most-able participants wanted training. The vertical line (M) indicates the mean. As may be expected and as is evident in Figure 3, the participants wanted training in the reading items and navigation items, such as getting around in unfamiliar places.

In Figure 4, this correlation is plotted between the Rasch measures for visual difficulty in performing an item versus the Rasch measures for the interest in training on that item. The items within the small oval in the lower-left quadrant are items that the participants found difficult and for which they wanted training: reading menus, reading small print on package labels, and reading newspaper or magazine articles. The items within the small oval in the upper-right quadrant are items that the participants found easy and for which they did not want training: keeping clean, getting around indoors in places they know, and getting dressed. The items within the larger oval are items for which the participants were less interested in training even though the tasks were difficult: playing sports, working on a favorite hobby, playing table and card games, using public transportation, doing yard work, going to the movies, going to spectator events, reading signs, and going out at night. These nine items may be weighted toward being more idiosyncratic or ambiguous and may not lend themselves easily to training. As we stated earlier, the item ordering for the difficulty question (items ranked from the easiest to the most difficult) was highly correlated with the item ordering for whether the participants wanted training ($R = .85$, $p \leq .01$). This correlation increased to $R = .98$ ($p \leq .01$) with the removal of the nine ineffective items shown in the larger oval. The rankings for the vision (subcomponent *b*) and training questions (subcomponent *c*) were also highly correlated ($R = .98$, $p \leq .01$).

The separation (S) and reliability (R) of the scales for subcomponents *a* through *c*, respectively, were high at $S = 3.91$ and $R = .94$, $S = 3.54$ and $R = .93$, and $S = 3.29$ and $R = .92$. (For a discussion of separation

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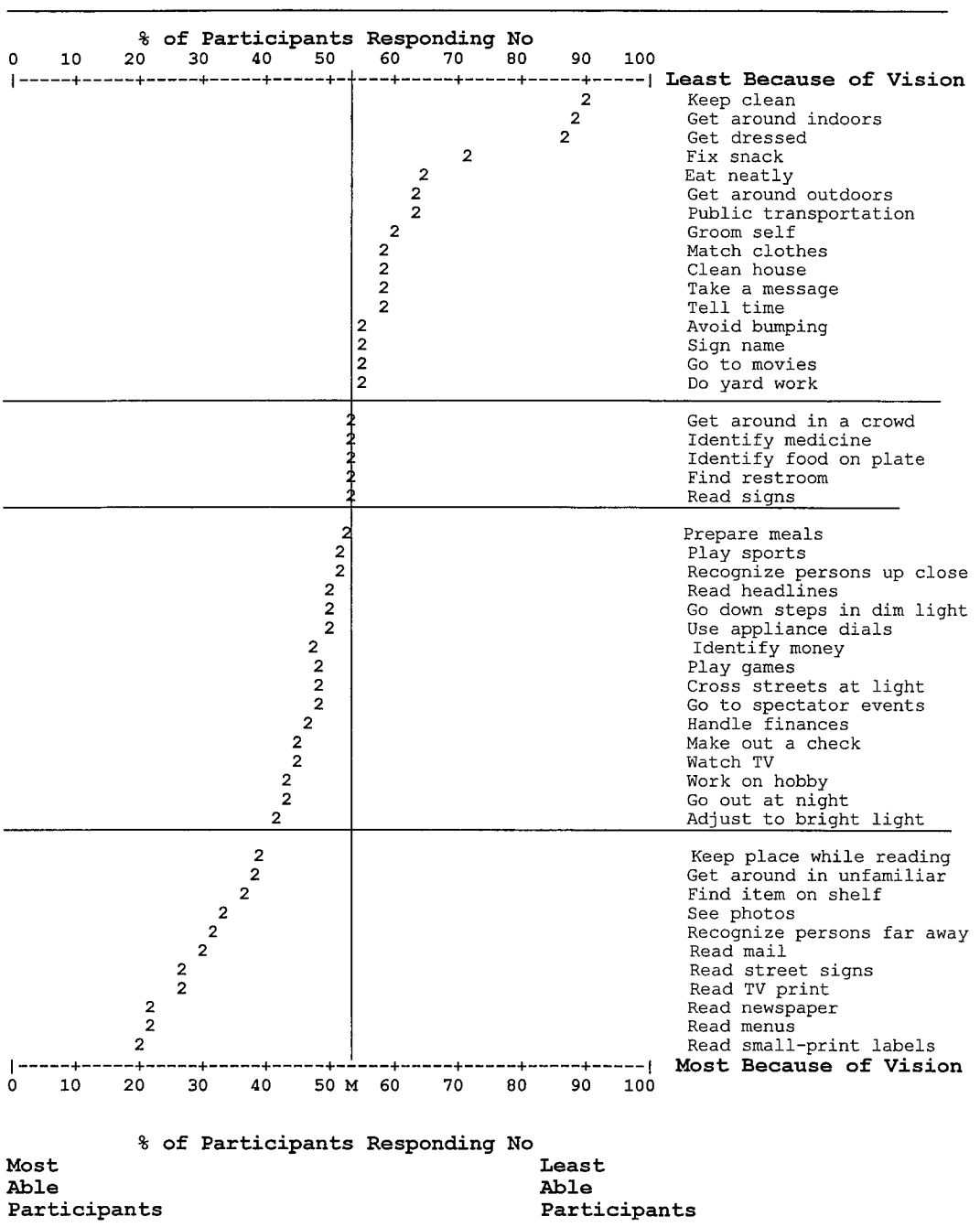


Figure 2. Ranking of responses to questionnaire subcomponent *b*: "Is it because of your vision?"

Do you want training?

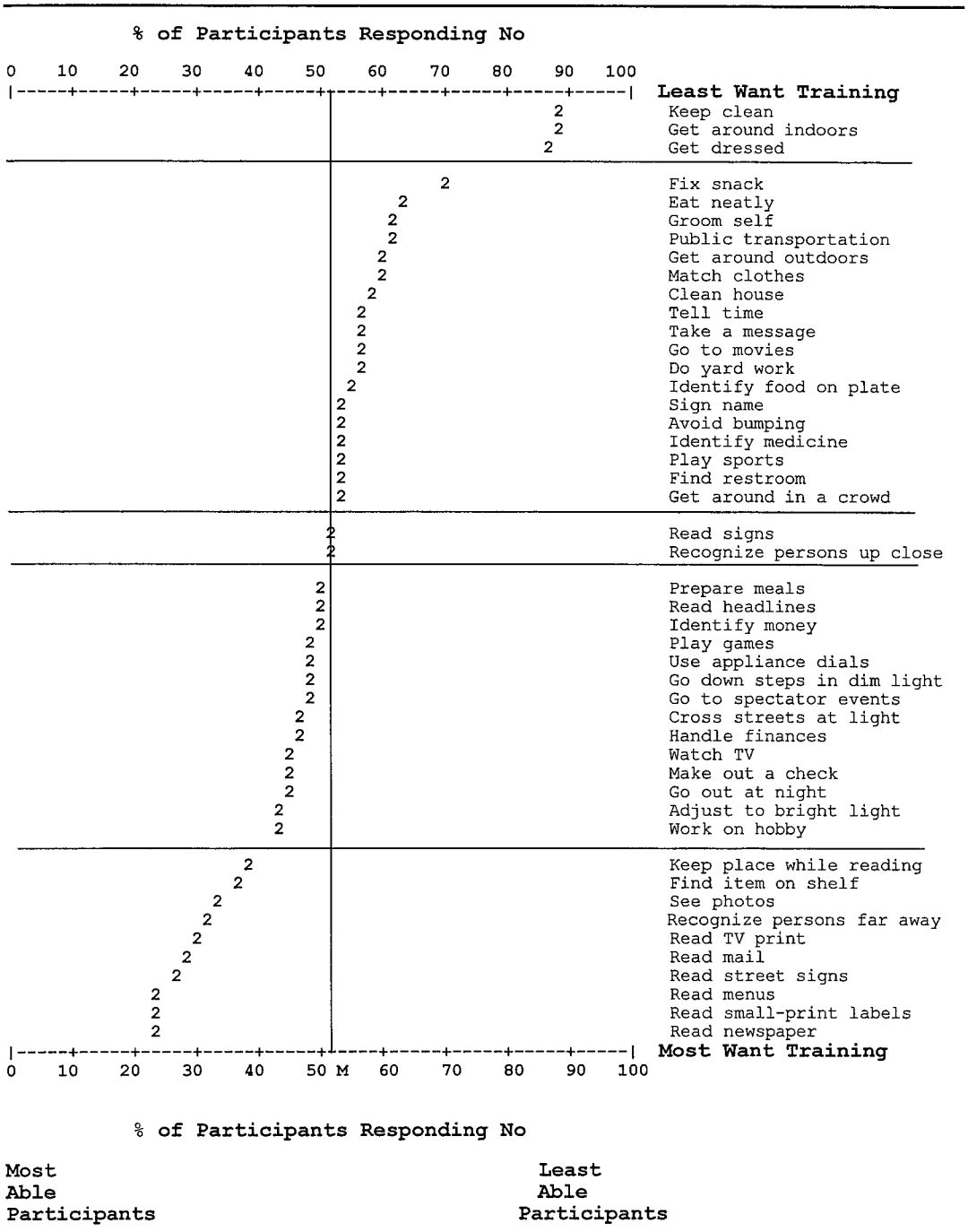


Figure 3. Ranking of responses to questionnaire subcomponent *c*: "Do you want training?"

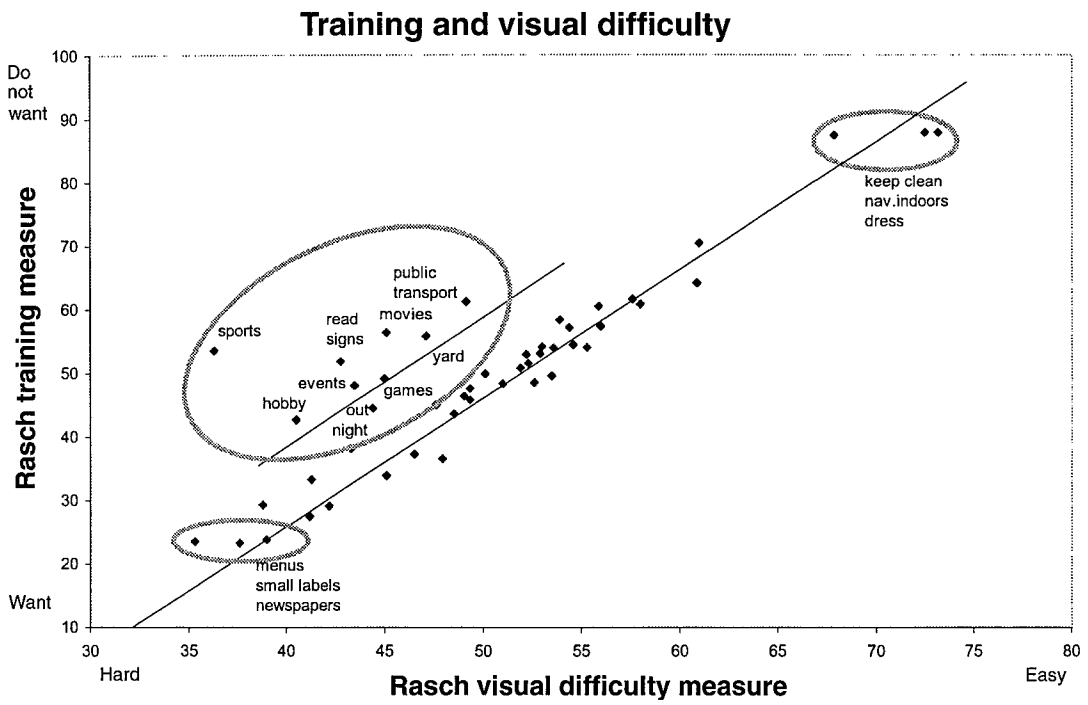


Figure 4. Rasch measures for the difficulty question, subcomponent *a,* plotted against the Rasch measures for the training question, subcomponent *c.*

and reliability in relation to the Rasch model, see Wright & Masters, 1982.) These values indicate that there was a high degree of discrimination between items on the scales and that the orderings were consistent across the participants.

A principal-components analysis on the item-measure residuals was performed on the training subcomponent (subcomponent *c*) of the questionnaire. The analysis discriminated two dimensions with positive and negative factor loadings. One dimension was labeled "Reading" (15 items) and the other was labeled "Mobility" (18 items) because they were differentiated by resolution-type tasks and navigation-type tasks. Figures 5 and 6 show the item-ordering scales for these two dimensions, Reading and Mobility, respectively.

Figure 5 presents the distribution of responses of the 117 participants from those reading items for which training was not requested (top) to those reading items for which training was requested (bottom). As the cascade of numbers descends to the left, it indicates items for which even the most-able participants wanted reading training. Vertical lines indicate 2 *SD* and the mean (M). Data were analyzed for 96 participants and 15 items. There were 6 participants who responded "No, I do not want training" to all the reading-oriented questions and 15 participants who responded "Yes, I do want training" to all these questions. Items with a mean response of 2 (meaning "No, I do not want training") or approximately 2 included making out a check, handling finances, watching television,

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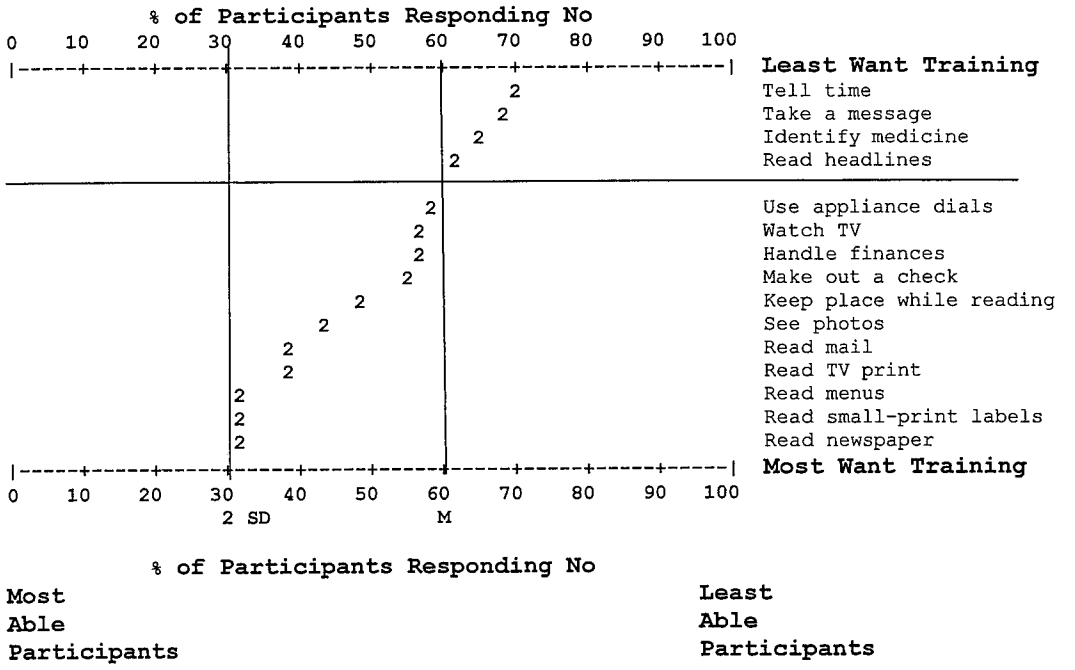


Figure 5. The scale for interest in reading training generated by the Rasch analysis.

using appliance dials, and reading headlines.

Figure 6 demonstrates the distribution of responses of the 117 participants from those mobility items for which training was the least requested (top) to those for which training was the most requested (bottom). A no response, "I do not want training," was coded 2. As the cascade of numbers in Figure 6 descends to the left, it indicates items for which even the most-able participants wanted mobility training. The vertical line (M) indicates the mean. Data were analyzed for 109 participants and 18 items because there were 6 participants who responded "No, I do not want training" to all the mobility-oriented questions and 2 participants who responded "Yes, I do want training" to all these questions. Items with a mean response of 2 or approximately 2 included preparing meals, recognizing persons up close, getting around in a crowd, and finding public restrooms.

In Figure 7, the Rasch training measure for reading is plotted versus the Rasch training measure for mobility for each participant. The two measures are significantly correlated, but at a lower level ($R = .53, p \leq .01$) than was shown between subcomponents. The participants whose plots are located above the diagonal axis were more interested in receiving mobility training than receiving reading training. Those whose plots are located below the diagonal axis were more interested in receiving reading training than receiving mobility training.

The Rasch analysis for subcomponent *d*, the level of independence, yielded a lack of variability when each possibility was

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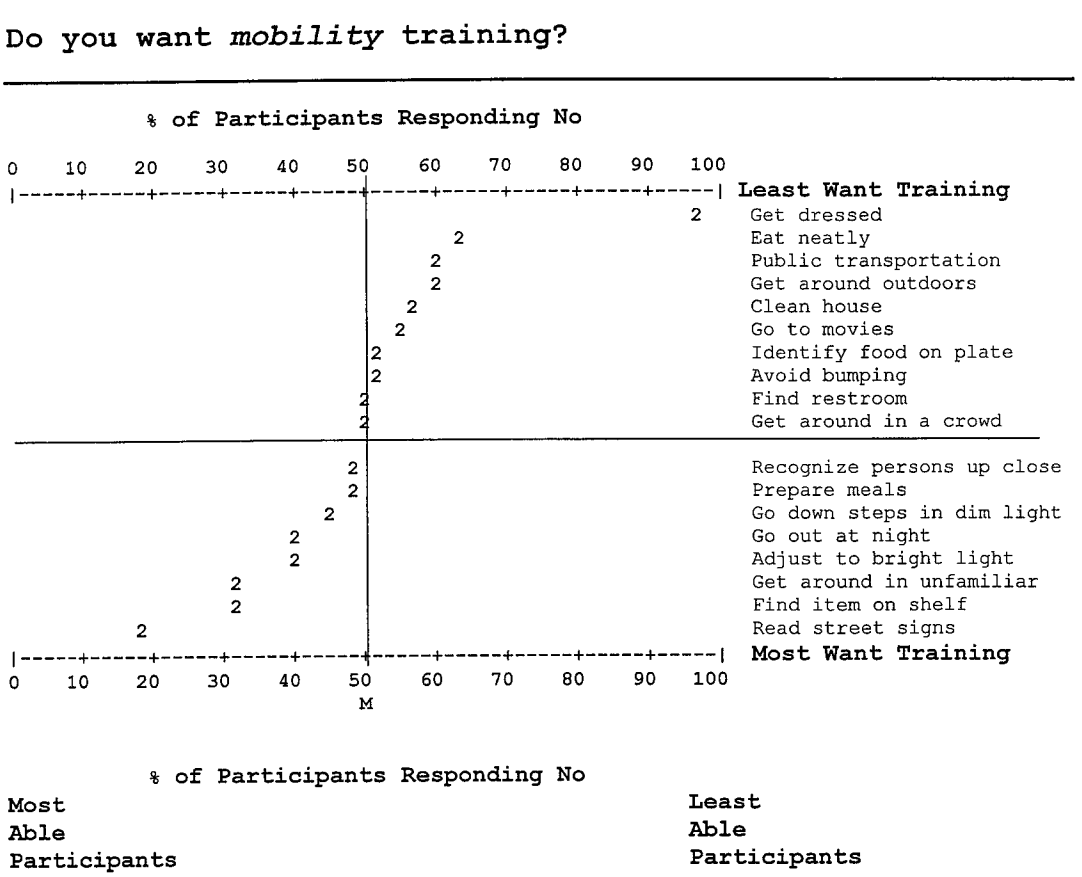


Figure 6. The scale for interest in mobility training generated by the Rasch analysis.

analyzed separately. Most participants did not use a low vision device; however, 19 participants reported using a low vision device for reading tasks (for example, reading menus, small print on package labels, newspapers, or magazine articles). An analysis of this question will be more important in the postrehabilitation analysis, once the participants have been provided with devices and given training in their use. The participants reported obtaining help from others on only 9 of the 48 tasks mentioned on the questionnaire: going out at night, finding public restrooms, making out a check, handling finances, reading signs, reading small print on package labels, reading mail, getting around in unfamiliar places, and reading menus.

Discussion

Limitations of the Research

Overall, there was a high level of correspondence between the level of difficulty of a task and the participants' interest in receiving training. However, on nine items, there were incongruent findings between difficulty and training (see Figure 4). Although the participants considered these items difficult, they were not interested in receiving training in these tasks. The participants may have perceived these items to be less relevant to their lives and their overall ability to function and survive in everyday life. They may also have perceived that training in activities, such as yard work, was out of the purview of a vi-

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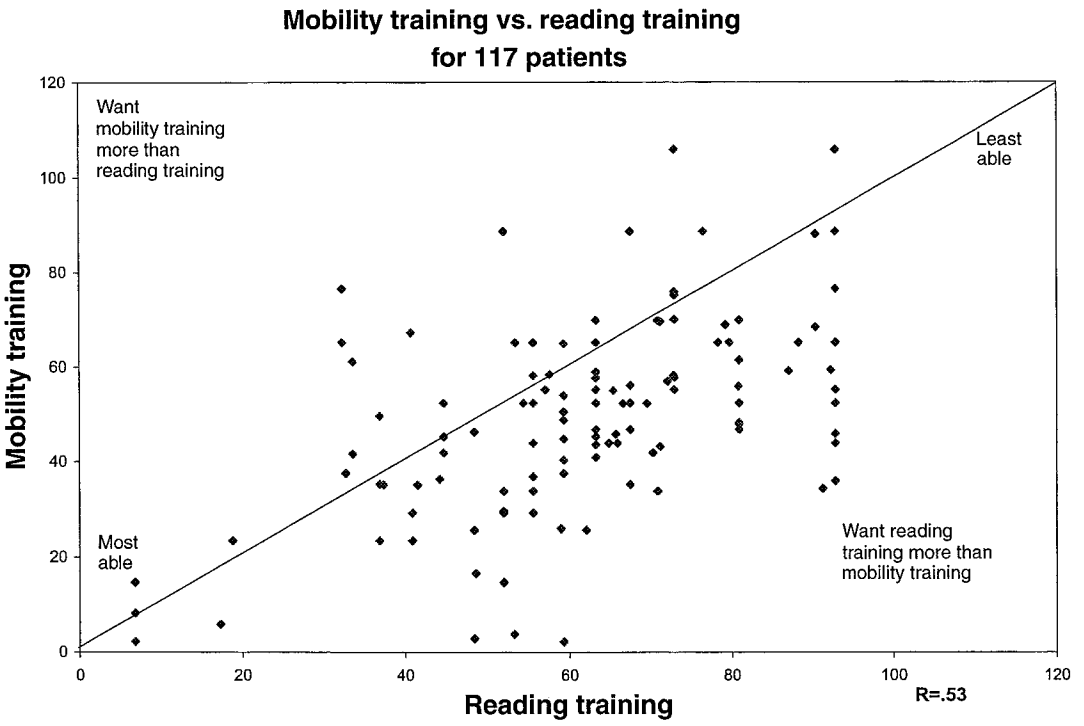


Figure 7. The Rasch reading-training measure versus the Rasch mobility-training measure.

sion rehabilitation program. It may be that the nine items with the incongruent findings could be eliminated from the questionnaire. However, it is prudent to wait for the postrehabilitation analyses to be completed. Items that were considered too easy could also be eliminated from the questionnaire. In the long term, these types of tasks may also be eliminated from curricula in low vision rehabilitation.

The item "reading street signs and store names" generated a different response profile from the generic question for reading signs (see Figure 1). Perhaps the participants can assess their degree of difficulty more accurately when they are provided with a specific instance or example of a task instead of a general question. The generic item "reading signs" could be confused with simply recognizing signs (using color, a gross logo, or other graphic cues) and thus could be rated as less difficult than "reading street signs and store names."

Another seeming discrepancy is that although the participants most commonly rated reading small print on labels as extremely difficult and wanted rehabilitation training for it, they scored identifying medicine as only slightly difficult and did not want training for it (see Figure 5). The two items may seem to be similar, in that medicines likely have small-print labels; however, medicines come in different formulations, shapes, and sizes and sometimes in potentially identifiable containers. Tactile variation may explain the difference in responses between these two items.

Implications for Practice

The results show that this questionnaire is an effective instrument for strati-

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fying and measuring visual difficulties in individuals who receive low vision rehabilitation. The Rasch analytic technique was critical in the development of a reliable yardstick with which to measure a participant's progress after participation in a program. Consistent with Massof's (1998) findings that recreational reading requires the most visual ability and self-care requires the least, we found that reading small print on package labels and reading newspapers were the most difficult tasks for the participants in our study and that keeping clean and getting dressed were among the least difficult tasks. The aim of Massof's study was to develop a hierarchy of importance, before persons undergo low-vision rehabilitation, for achieving daily goals independently. Our study showed that getting around in familiar places, such as the home, was not difficult for the participants. This finding is consistent with the finding by Turano et al. (1999), which used Rasch analysis, and that "moving about in the home" required the least visual ability for participants with RP. Use of the Veterans Affairs Low Vision Visual Functioning Questionnaire and tracking whether a participant prefers reading or mobility training (see Figure 7) would provide a clearer direction for planning a customized rehabilitation program.

Implications for Research

In future research, we will assess the reliability of the rank ordering of items after rehabilitation. This will be a further test of the integrity of the difficulty scale. The work by Turano et al. (1999) to define a mobility scale for participants with peripheral vision loss and the two dimensions of reading and mobility defined in our study emphasize the need to develop interval scales that are based on the predominant visual profiles in study groups. In this study, the participants' interest in receiving training was differentiated on the basis of their needs. Some participants preferred to receive training in reading-related items, while others tended to prefer training on items that focused on mobility (see Figure 7). These preferences were related to the loss of central vision versus the loss of peripheral vision, respectively. However, this is a point that is beyond the scope of this article and needs to be explored in greater depth in subsequent articles dealing with this measure. Future rehabilitation programs may want to consider including a choice of either reading- or mobility-focused tracks to make better use of the participants' and specialists' time.

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